



BETS

Institute of Technology

Network Protocols Reference Assignment

Top 100 Protocols — Layer, Port, Transport & Usage

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Network Protocols Reference Table

The table below lists **100 network protocols** commonly encountered in networking and cybersecurity. Each entry shows the OSI layer the protocol operates at, its standard port number, whether it uses TCP or UDP transport, whether it provides built-in encryption, and a brief description of its primary use.

Protocols are ordered starting from the well-known application-layer protocols, then moving through transport, network, and data-link layer protocols, and finishing with tunnelling and VPN protocols.

#	Protocol	Layer	Port Number	TCP / UDP	Encryption	Usage
1	HTTP	Application	80	TCP	No	Web browsing (unsecured)
2	HTTPS	Application	443	TCP	Yes	Secure web browsing (TLS)
3	FTP	Application	20/21	TCP	No	File transfer between hosts
4	FTPS	Application	990	TCP	Yes	FTP over TLS/SSL
5	SFTP	Application	22	TCP	Yes	Secure file transfer over SSH
6	SSH	Application	22	TCP	Yes	Encrypted remote shell access
7	Telnet	Application	23	TCP	No	Remote login (unencrypted)
8	SMTP	Application	25	TCP	No	Sending email
9	SMTPS	Application	465	TCP	Yes	Secure email sending (TLS)
10	SMTP STARTTLS	Application	587	TCP	Yes	Email submission with STARTTLS
11	POP3	Application	110	TCP	No	Retrieving email from server
12	POP3S	Application	995	TCP	Yes	Secure POP3 over TLS
13	IMAP	Application	143	TCP	No	Email access and management
14	IMAPS	Application	993	TCP	Yes	Secure IMAP over TLS
15	DNS	Application	53	TCP/UDP	No	Domain name resolution
16	DNSSEC	Application	53	TCP/UDP	Yes	Authenticated DNS lookups
17	DHCP	Application	67/68	UDP	No	Automatic IP address assignment
18	TFTP	Application	69	UDP	No	Simple file transfer, no auth
19	SNMP v1/v2	Application	161	UDP	No	Network device monitoring
20	SNMP v3	Application	161	UDP	Yes	Secure network device monitoring
21	SNMP Trap	Application	162	UDP	No	Alerts from network devices
22	NTP	Application	123	UDP	No	Time synchronisation
23	NTPv4	Application	123	UDP	Yes	Authenticated time sync
24	LDAP	Application	389	TCP	No	Directory services lookup
25	LDAPS	Application	636	TCP	Yes	Secure LDAP over TLS
26	RDP	Application	3389	TCP	Yes	Remote desktop access (Windows)
27	VNC	Application	5900	TCP	No	Remote desktop sharing
28	SIP	Application	5060	TCP/UDP	No	VoIP call signalling

29	SIPS	Application	5061	TCP	Yes	Secure VoIP signalling (TLS)
30	H.323	Application	1720	TCP	No	VoIP and video conferencing
31	RTP	Application	5004	UDP	No	Real-time audio/video transport
32	RTSP	Application	554	TCP/UDP	No	Streaming media control
33	RTMP	Application	1935	TCP	No	Live video streaming (Flash)
34	XMPP	Application	5222	TCP	No	Instant messaging (Jabber)
35	IRC	Application	6667	TCP	No	Internet Relay Chat
36	Kerberos	Application	88	TCP/UDP	Yes	Network authentication
37	RADIUS	Application	1812	UDP	Yes	Remote user authentication
38	TACACS+	Application	49	TCP	Yes	Device admin authentication
39	SYSLOG	Application	514	UDP	No	System log messages
40	Syslog TLS	Application	6514	TCP	Yes	Secure syslog over TLS
41	RSYNC	Application	873	TCP	No	File synchronisation utility
42	NNTP	Application	119	TCP	No	Usenet newsgroup access
43	WHOIS	Application	43	TCP	No	Domain registration lookup
44	Finger	Application	79	TCP	No	User information lookup
45	GOPHER	Application	70	TCP	No	Pre-web document retrieval
46	MQTT	Application	1883	TCP	No	IoT lightweight messaging
47	MQTTs	Application	8883	TCP	Yes	Secure MQTT over TLS
48	AMQP	Application	5672	TCP	No	Message queue protocol
49	STOMP	Application	61613	TCP	No	Text-oriented messaging
50	CoAP	Application	5683	UDP	No	Constrained IoT protocol
51	HTTP/2	Application	443	TCP	Yes	Multiplexed web browsing
52	HTTP/3	Application	443	UDP	Yes	Web over QUIC transport
53	QUIC	Transport	443	UDP	Yes	Low-latency encrypted transport
54	TLS	Session	443	TCP	Yes	Encryption layer for protocols
55	DTLS	Session	Various	UDP	Yes	TLS for datagram protocols
56	SSL	Session	443	TCP	Yes	Older encryption (deprecated)
57	TCP	Transport	—	TCP	No	Reliable connection-oriented comms
58	UDP	Transport	—	UDP	No	Fast connectionless comms
59	SCTP	Transport	—	SCTP	No	Multi-stream reliable transport
60	DCCP	Transport	—	UDP	No	Datagram congestion control
61	IP v4	Network	—	N/A	No	Internet addressing (32-bit)
62	IP v6	Network	—	N/A	No	Internet addressing (128-bit)
63	ICMP	Network	—	N/A	No	Error reporting and ping
64	ICMPv6	Network	—	N/A	No	IPv6 error and discovery msgs

65	IGMP	Network	—	N/A	No	Multicast group management
66	OSPF	Network	89	N/A	No	Link-state routing protocol
67	RIP v1/v2	Network	520	UDP	No	Distance-vector routing
68	RIPng	Network	521	UDP	No	RIP for IPv6 networks
69	EIGRP	Network	88	N/A	No	Cisco advanced distance-vector
70	BGP	Application	179	TCP	No	Inter-AS internet routing
71	IS-IS	Network	—	N/A	No	Link-state routing (ISPs)
72	MPLS	Data Link	—	N/A	No	Label-based packet forwarding
73	ARP	Data Link	—	N/A	No	IP-to-MAC address resolution
74	RARP	Data Link	—	N/A	No	MAC-to-IP reverse lookup
75	NDP	Network	—	N/A	No	IPv6 neighbour discovery
76	STP	Data Link	—	N/A	No	Prevents Layer-2 loops
77	RSTP	Data Link	—	N/A	No	Rapid spanning tree
78	LACP	Data Link	—	N/A	No	Link aggregation control
79	LLDP	Data Link	—	N/A	No	Neighbour device discovery
80	CDP	Data Link	—	N/A	No	Cisco device discovery
81	PPP	Data Link	—	N/A	No	Point-to-point WAN links
82	PPPoE	Data Link	—	N/A	No	PPP over Ethernet (DSL)
83	HDLC	Data Link	—	N/A	No	WAN serial link framing
84	Frame Relay	Data Link	—	N/A	No	WAN packet-switched network
85	ATM	Data Link	—	N/A	No	Cell-based WAN transport
86	Ethernet	Data Link	—	N/A	No	LAN wired communication
87	Wi-Fi 802.11	Data Link	—	N/A	No	Wireless LAN communication
88	WPA3	Data Link	—	N/A	Yes	Secure wireless authentication
89	Bluetooth	Data Link	—	N/A	No	Short-range wireless comms
90	Zigbee	Data Link	—	N/A	No	Low-power IoT wireless
91	IPSec	Network	500	UDP	Yes	VPN and encrypted tunnels
92	IKEv2	Network	500	UDP	Yes	VPN key exchange
93	L2TP	Data Link	1701	UDP	No	VPN tunnelling protocol
94	GRE	Network	47	N/A	No	Generic encapsulation tunnel
95	VXLAN	Data Link	4789	UDP	No	Virtual LAN over IP network
96	WSUS	Application	8530	TCP	No	Windows update distribution
97	SMB	Application	445	TCP	No	Windows file/printer sharing
98	NetBIOS	Application	137-139	TCP/UDP	No	Legacy Windows name service
99	ISAKMP	Application	500	UDP	Yes	Key management for IPSec
100	OpenVPN	Application	1194	UDP	Yes	SSL-based VPN solution

